

Lab 4

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7-Segment Display

Creating the Truth Table for the 7-segment display by creating (0-9). After 9, the pattern should repeat from 0-5.

Table 1.

i_m3	i_m2	i_m1	i_m0	o_seg[7..0]
0	0	0	0	11000000
0	0	0	1	11111001
0	0	1	0	10100100
0	0	1	1	10101000
0	1	0	0	10011001
0	1	0	1	10011001
0	1	1	0	10010010
0	1	1	1	10000010
1	0	0	0	10000111
1	0	0	1	10000000
1	0	1	0	10010000
1	0	1	1	1-----
1	1	0	0	1-----
1	1	0	1	1-----
1	1	1	0	1-----
1	1	1	1	1-----

This is the table I tried initially but there was some mistakes and I forgot to repeat the pattern, so my segment display was wack!!

Table 2.

-

1

0

Collapse Duplicated Rows

Show All Rows

16 of 16 rows shown

i_m3	i_m2	i_m1	i_m0	o_seg[7..0]
0	0	0	0	11000000
0	0	0	1	11111001
0	0	1	0	10100100
0	0	1	1	10011001
0	1	0	0	10011001
0	1	0	1	10001010
0	1	1	0	10000010
0	1	1	1	10000000
1	0	0	0	11101111
1	0	0	1	10111111
1	0	1	0	10101111
1	0	1	1	10111111
1	1	0	0	10101111
1	1	0	1	10111111
1	1	1	0	10111111
1	1	1	1	10000000

This is the corrected table and in image 1, you will see what it looks like as a circuit. NAND and NOR gates.

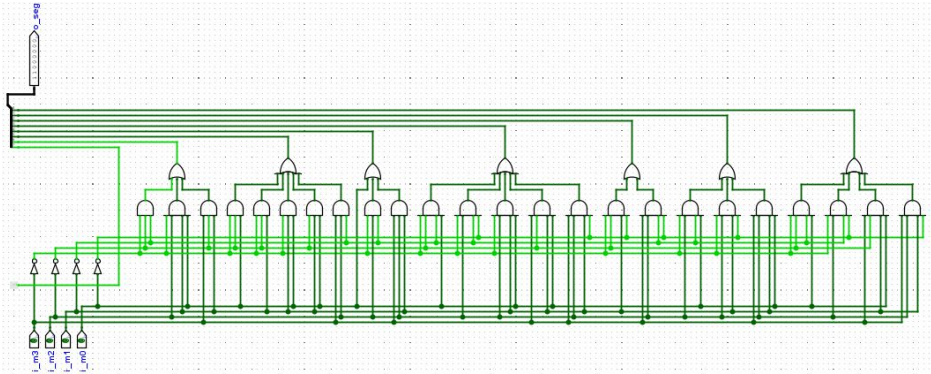
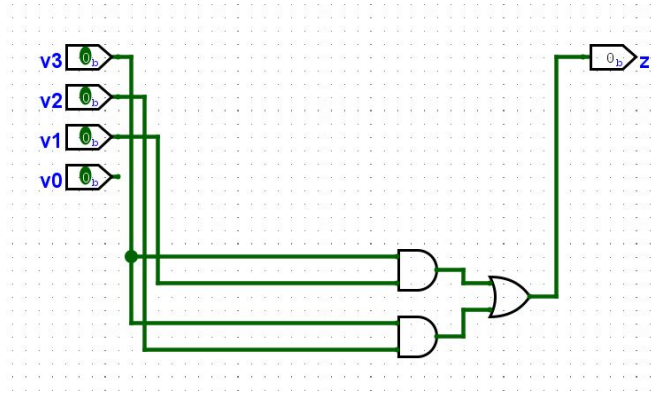


Image 1.

Comparator



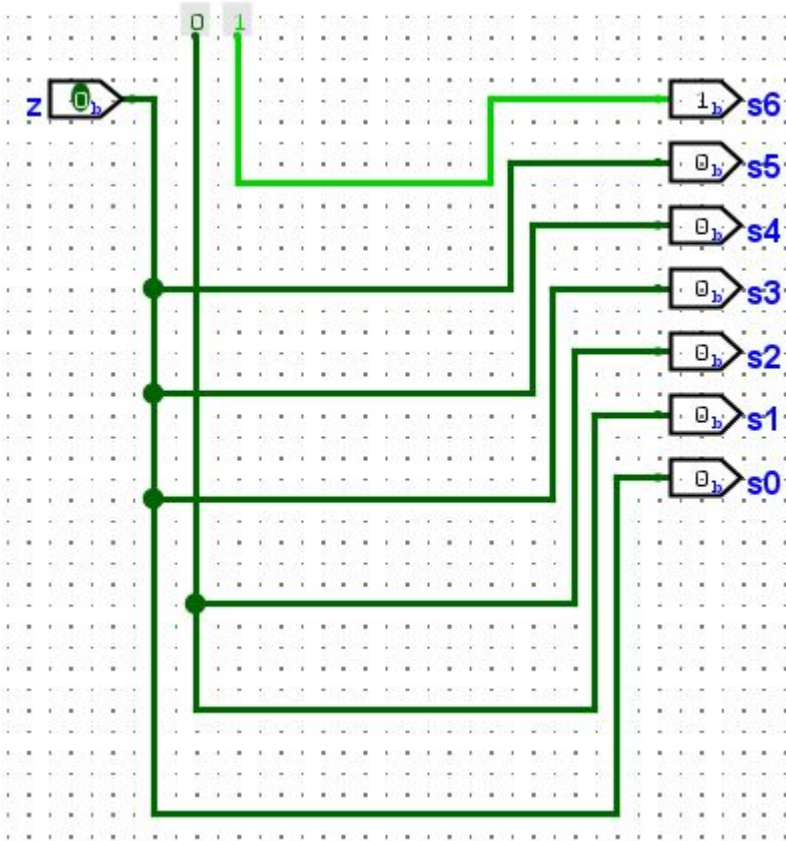
This is the comparator it's working with the 7-seg decoder and it's helping compare the two voltages between circuit b and the 7 segment decoder. This will measure the high and low of the circuits it's connected to. (I think this also shows whether or not certain numbers will show up or not since it can't go above #14. This is letting the system know if operand A or operand B are operating at the same time/what number shall display on each one.)

16 of 16 rows shown

v3	v2	v1	v0	z
0	0	0	0	0
0	0	0	1	0
0	0	1	0	0
0	0	1	1	0
0	1	0	0	0
0	1	0	1	0
0	1	1	0	0
0	1	1	1	0
1	0	0	0	0
1	0	0	1	0
1	0	1	0	1
1	0	1	1	1
1	1	0	0	1
1	1	0	1	1
1	1	1	0	1
1	1	1	1	1

This is the truth table for the comparator that's connected to 7 segment decoder and circuit B.

For circuit B, the z will render if it will be a 0 or 1

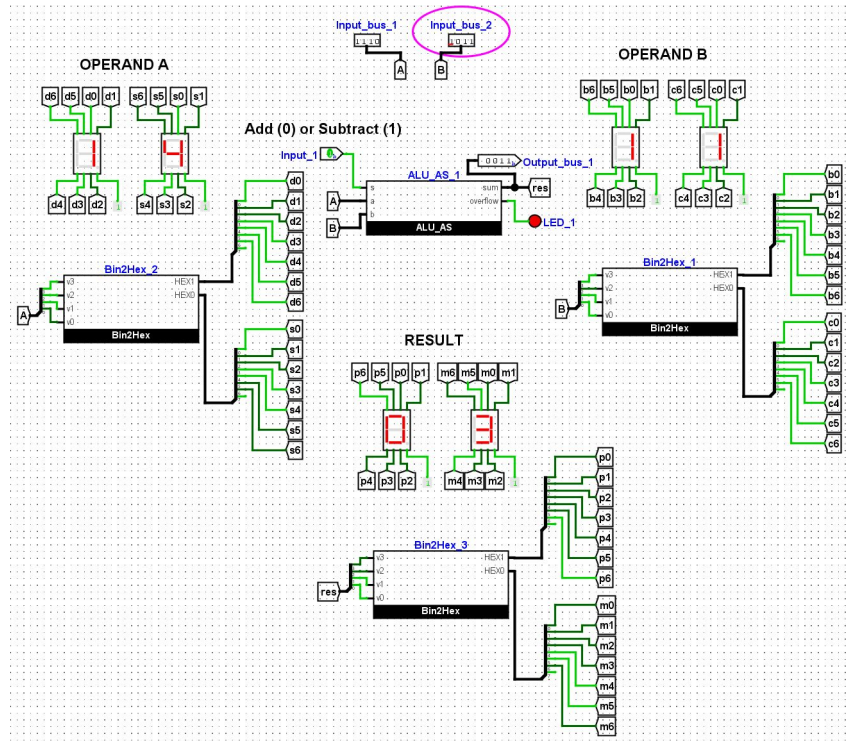


2 of 2 rows shown

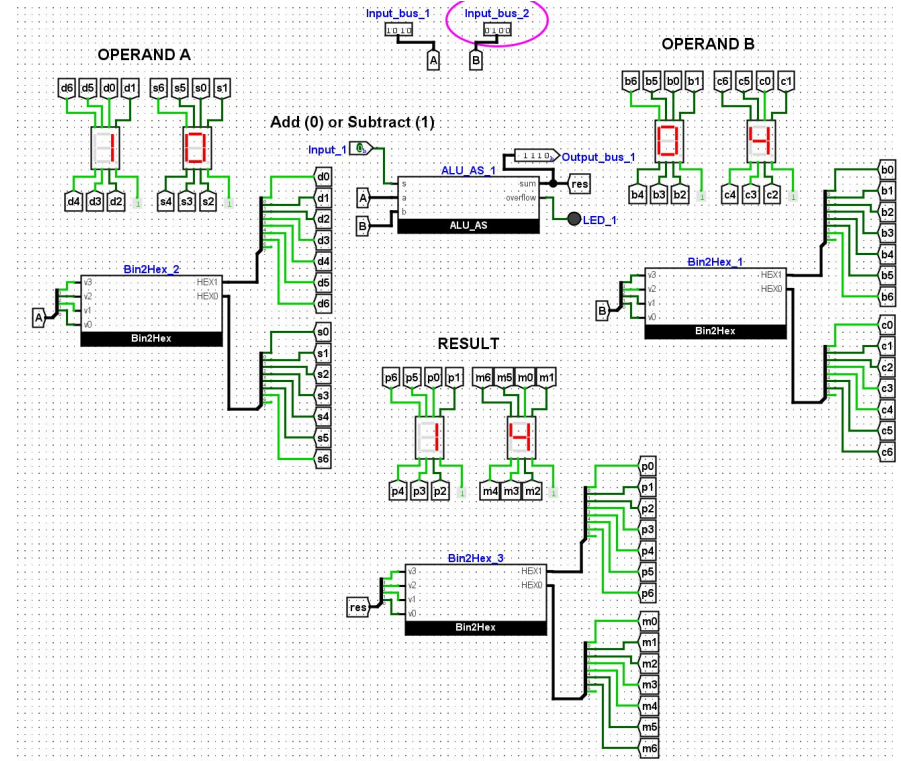
z	s6	s5	s4	s3	s2	s1	s0
0	1	0	0	0	0	0	0
1	1	1	1	1	0	0	1

This is circuit B, and its truth table. It will show zero or one to indicate whether it's on or off depending on which number the seven segment result is going to display.

Testing to make sure all the components are working in order to add and subtract properly.



Subtracting 14-11 and getting 3 as a result.



Adding 10 + 4 and getting 14 as a result.